Stewart, Calculus: Early Transcendentals

Section 15.8, Exercise 26: a cylinder cuts a sphere

Problem

Use cylindrical coordinates.

(a) Find the volume of the solid that the cylinder $r = a\cos\theta$ cuts out of the sphere of radius a centered at the origin.

(b) Illustrate the solid of part (a) by graphing the sphere and the cylinder on the same screen.

Symbols used in this solution

- E is the solid whose volume we want: the set of points that lie inside the ball $x^2 + y^2 + z^2 \le a^2$ and also inside the cylinder $r \le a\cos\theta$.
- r, θ, z are cylindrical coordinates: $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, and $r = \sqrt{x^2 + y^2} \ge 0$.
- dV is the volume element. In cylindrical coordinates $dV = r\,dz\,dr\,d\theta$. The factor r is the Jacobian, and it is the reason a cylindrical volume integral is not simply $\iiint dz\,dr\,d\theta$.
- a is the radius of the sphere. It is a positive constant, and the answer is a formula in a. The numeric checks use $a = 1$.

Part (a): the exact volume

Step 1. The cylinder is a circle in the plane of constant z

Start from the polar equation and multiply by r:

$$r = a\cos\theta \quad\Longleftrightarrow\quad r^2 = a\,r\cos\theta \quad\Longleftrightarrow\quad x^2 + y^2 = ax. \tag{1}$$

Multiply by r is safe because $r \ge 0$, and $r = 0$ gives the single point $(0, 0)$, which the circle already contains. Complete the square in x:

$$x^2 - ax + y^2 = 0 \quad\Longleftrightarrow\quad \left(x - \tfrac{a}{2}\right)^2 + y^2 = \left(\tfrac{a}{2}\right)^2. \tag{2}$$

Equation (2) is a circle of radius $a/2$ with center $(a/2, 0)$. In space it is a cylinder of radius $a/2$ whose axis is the line $x = a/2$, $y = 0$, parallel to the z-axis. The cylinder touches the z-axis at the origin, and it lies entirely in the half-space $x \ge 0$.

The solid E is the part of the ball that is inside this cylinder:

$$E = \left\{(x,y,z) : x^2+y^2+z^2 \le a^2,\ \left(x-\tfrac{a}{2}\right)^2 + y^2 \le \left(\tfrac{a}{2}\right)^2\right\}. \tag{3}$$

Step 2. The range of θ

The description $0 \le r \le a\cos\theta$ needs $a\cos\theta \ge 0$, that is $\cos\theta \ge 0$. One full turn gives $\cos\theta \ge 0$ exactly on

$$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}. \tag{4}$$

On that range $\cos\theta$ falls from 0 at $\theta = \pm\pi/2$ to 1 at $\theta = 0$, so the radius $r = a\cos\theta$ grows from 0 to a. This matches the circle (2): it starts at the origin, reaches $(a,0)$ at $\theta = 0$, and returns to the origin.

The projection D of E on the plane $z = 0$ is exactly the disc of radius $a/2$ in (2). Its area is

$$\text{area}(D) = \frac{1}{2}\int_{-\pi/2}^{\pi/2} \big(a\cos\theta\big)^2 d\theta = \frac{a^2}{2}\cdot\frac{\pi}{2} = \frac{\pi a^2}{4}, \tag{5}$$

which is $\pi(a/2)^2$, as it must be.

Step 3. The z-limits and the order of integration

Fix a point (r,θ) with $0 \le r \le a\cos\theta$, that is a point of the disc D. The vertical line through that point meets the ball in the segment

$$-\sqrt{a^2-r^2} \ \le\ z\ \le\ \sqrt{a^2-r^2}, \tag{6}$$

because $z^2 \le a^2 - r^2$. The cylinder puts no further condition on z: it covers every z above the disc. Therefore every vertical line meets E in one closed segment, and E is z-simple over D. This is why we may write

$$V = \iiint_E dV = \int_{-\pi/2}^{\pi/2}\int_0^{a\cos\theta}\int_{-\sqrt{a^2-r^2}}^{\sqrt{a^2-r^2}} r\,dz\,dr\,d\theta. \tag{7}$$

The order $dz\,dr\,d\theta$ is legal for exactly this reason: the inner integration runs along a line, from the lower sheet of the sphere to the upper sheet, and the outer two integrations then run over the disc D in polar form.

The radius $a^2 - r^2$ is non-negative on D, because $r \le a\cos\theta \le a$. On the rim of the disc $r = a\cos\theta$, so the height is

$$\sqrt{a^2-r^2} = \sqrt{a^2 - a^2\cos^2\theta} = a\,|\sin\theta|, \tag{8}$$

and the solid has zero thickness there. That is the pointed rim seen in Figure 2.

Step 4. The inner two integrations

The integrand does not depend on z, so the innermost integral is

$$\int_{-\sqrt{a^2-r^2}}^{\sqrt{a^2-r^2}} dz = 2\sqrt{a^2-r^2}. \tag{9}$$

Now integrate in r. Put $u = a^2 - r^2$, $du = -2r\,dr$. Then

$$\int 2r\sqrt{a^2 - r^2}\,dr = -\frac{2}{3}\big(a^2 - r^2\big)^{3/2}, \tag{10}$$

because the derivative of $-\frac{2}{3}(a^2 - r^2)^{3/2}$ is $-\frac{2}{3}\cdot\frac{3}{2}(a^2 - r^2)^{1/2}(-2r) = 2r\sqrt{a^2 - r^2}$. Evaluate from $r = 0$ to $r = a\cos\theta$:

$$\int_0^{a\cos\theta} 2r\sqrt{a^2 - r^2}\,dr = \frac{2}{3}\Big(a^3 - \big(a^2 - a^2\cos^2\theta\big)^{3/2}\Big) = \frac{2a^3}{3}\Big(1 - |\sin\theta|^3\Big). \tag{11}$$

The absolute value is essential. On part of the range $\sin\theta < 0$, and $\big(a^2\sin^2\theta\big)^{3/2} = a^3|\sin\theta|^3$, not $a^3\sin^3\theta$. Dropping the absolute value doubles the negative half of the answer and gives the wrong value.

Step 5. The θ integration

Insert (11) into (7):

$$V = \frac{2a^3}{3}\int_{-\pi/2}^{\pi/2}\Big(1 - |\sin\theta|^3\Big)d\theta. \tag{12}$$

The integrand is even in θ, so we double the integral over $[0, \pi/2]$:

$$\int_{-\pi/2}^{\pi/2}\Big(1 - |\sin\theta|^3\Big)d\theta = 2\int_0^{\pi/2}\big(1 - \sin^3\theta\big)d\theta = 2\Big(\frac{\pi}{2} - \frac{2}{3}\Big) = \pi - \frac{4}{3}. \tag{13}$$

The standard reduction used here is

$$\int_0^{\pi/2}\sin^3\theta\,d\theta = \int_0^{\pi/2}\big(1 - \cos^2\theta\big)\sin\theta\,d\theta = \Big[-\cos\theta + \frac{\cos^3\theta}{3}\Big]_0^{\pi/2} = 1 - \frac{1}{3} = \frac{2}{3}. \tag{14}$$

Therefore

$$V = \frac{2a^3}{3}\Big(\pi - \frac{4}{3}\Big) = \frac{2a^3(3\pi - 4)}{9}. \tag{15}$$

Step 6. Checks

Table 1 collects the checks. The exact value was obtained twice with `sympy`: once from the cylindrical integral (7), and once from the complementary route that subtracts from the ball the half of the ball with $\cos\theta < 0$ and the crescent $r > a\cos\theta$. A third exact route integrates z first and then the disc in the order $dr\,d\theta$; it gives the same closed form. The numbers are independent quadratures and Monte Carlo estimates at $a = 1$.

Two classical formulae confirm the answer from outside. (i) The fraction of the ball that the solid occupies is

$$\frac{V}{\frac{4}{3}\pi a^3} = \frac{2a^3(3\pi - 4)/9}{4\pi a^3/3} = \frac{3\pi - 4}{6\pi} = \frac{1}{2} - \frac{2}{3\pi} \approx 0.287793, \tag{16}$$

so the offset cylinder takes less than one third of the ball. (ii) A cylinder of radius $a/2$ whose axis passes through the center of the ball would cut out the larger "lens"

$$\frac{4}{3}\pi a^3\Big[1 - \Big(1 - \tfrac{1}{4}\Big)^{3/2}\Big] = \frac{\pi a^3(8 - 3\sqrt{3})}{6} \approx 1.468091\,a^3, \tag{17}$$

Table 1: Independent checks of $V = 2a^3(3\pi - 4)/9$ at $a = 1$. The exact value is $1.2055062135043\ldots$

Method	Value at $a = 1$	Error
exact, `sympy` (three routes)	1.2055062135043066	0
2D Simpson rule in (θ, r)	1.205506174994	3.9×10^{-8}
Simpson rule for the x-slices	1.205506213504	8×10^{-15}
Monte Carlo in the box, 4×10^6 points	1.205127 ± 0.000918	3.8×10^{-4}
Monte Carlo inside the ball, 4×10^6	1.206906 ± 0.000949	1.4×10^{-3}

which passes through the same disc but uses the full length of the ball on both sides of that disc. Our solid has the same cross-section and the same radius $a/2$, but its axis is shifted by $a/2$, so it must be smaller. The numbers $1.205506 < 1.468091$ satisfy that requirement.

A useful sanity check of the geometry: the area of the cylinder wall inside the ball is $4a^2$, the area of the two spherical pieces is $2a^2(\pi - 2)$, and their sum is $2\pi a^2$, exactly half of the surface of the ball. The scripts verify these numbers exactly and from the plotted mesh.

$$\boxed{V = \frac{2a^3(3\pi - 4)}{9} = \frac{2a^3}{3}\Big(\pi - \frac{4}{3}\Big) \approx 1.205506\, a^3} \tag{18}$$

Part (b): graphing the sphere and the cylinder

Figure 1 puts the two surfaces on the same screen, and Figure 2 shows the solid that they bound.

Both surfaces are drawn from parametric formulas, so every plotted point satisfies its equation to machine precision (this is checked in `part2_step1.py`):

$$\text{sphere:}\quad x = a\sin\phi\cos\theta,\quad y = a\sin\phi\sin\theta,\quad z = a\cos\phi,\quad 0 \le \phi \le \pi; \tag{19}$$

$$\text{cylinder:}\quad x = \tfrac{a}{2}\big(1 + \cos\psi\big),\quad y = \tfrac{a}{2}\sin\psi,\quad z \text{ free.} \tag{20}$$

For the cylinder surface the identity $\big(\frac{a}{2}(1+\cos\psi)\big)^2 + \big(\frac{a}{2}\sin\psi\big)^2 = a^2\cos^2(\psi/2)$ shows that $\theta = \psi/2$, so the parameter ψ runs over one full turn while θ runs over $[-\pi/2, \pi/2]$.

What the picture shows:

- The cylinder stands on the disc D of radius $a/2$ centered at $(a/2, 0)$. The disc touches the y-axis at the origin.
- The sphere meets the cylinder along the red curve. That curve passes through $(a, 0, 0)$ and through both poles $(0, 0, \pm a)$.
- The solid E is the part of the ball on the same side of the cylinder wall as the positive x-axis. It is the "lemon" of Figure 2: two spherical pieces, one above the plane $z = 0$ and one below it, closed by the cylinder wall.
- The wall has zero height on the rim $r = a\cos\theta$ and full height $2a|\sin\theta|$ at the two poles, so the solid is pointed at $(a, 0, 0)$ and pinched at the poles. This is the geometric reason the height $\sqrt{a^2 - r^2}$ appears in the integral.

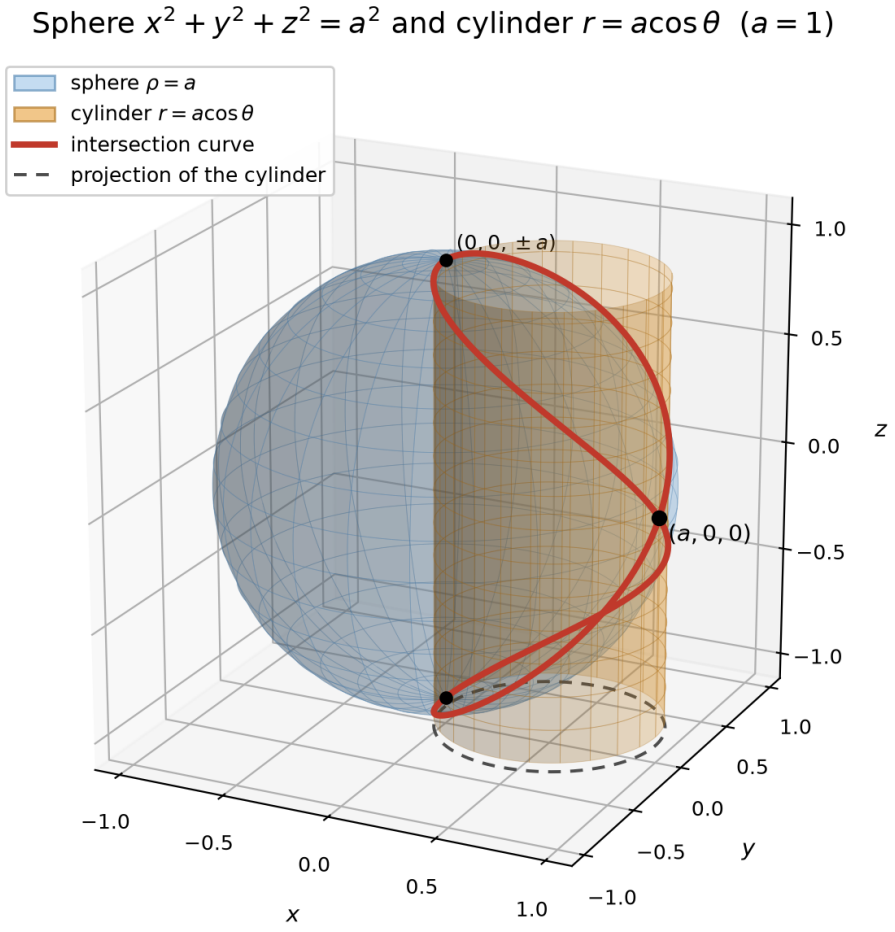


Figure 1: The sphere $\rho = a$ (blue, translucent) and the cylinder $r = a\cos\theta$ (orange, translucent) for $a = 1$. The cylinder is the circle of radius $a/2$ with center $(a/2, 0)$ extruded along z; it touches the z-axis at the origin and lies in $x \geq 0$. The red curve is the intersection of the two surfaces. The dashed grey circle is the projection of the cylinder on the plane $z = 0$.

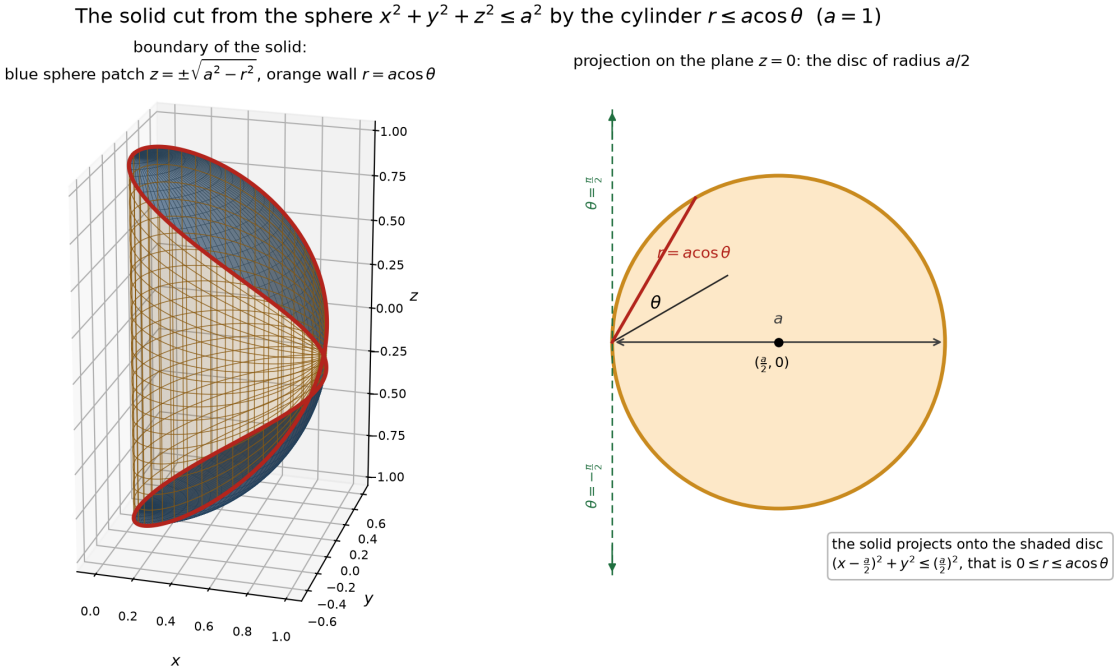


Figure 2: Left: the boundary of the solid, the blue spherical pieces $z = \pm\sqrt{a^2 - r^2}$ and the orange wireframe cylinder wall $r = a\cos\theta$, joined along the red intersection curve. Right: the projection on the plane $z = 0$. It is the disc of radius $a/2$, described in polar form by $0 \le r \le a\cos\theta$ with $-\pi/2 \le \theta \le \pi/2$. The red ray shows a typical θ and the radius $r = a\cos\theta$.

The two figures can also be checked against the numbers of part (a). Put $\theta = \psi/2$ in the wall parametrization. The wall is a cylinder of radius $a/2$ about the line $x = a/2$, $y = 0$, so a thin vertical strip of wall has width $\frac{a}{2}d\psi = a\,d\theta$ and height $2a|\sin\theta|$. Hence the wall inside the ball has area

$$\int_{-\pi/2}^{\pi/2} 2a|\sin\theta| \cdot a\,d\theta = 2a^2 \int_{-\pi/2}^{\pi/2} |\sin\theta|\,d\theta = 2a^2 \cdot 2 = 4a^2. \tag{21}$$

The two spherical pieces have total area $2a^2(\pi - 2)$; the scripts obtain this from the surface element of the sphere. The boundary drawn in Figure 2 therefore has total area $4a^2 + 2a^2(\pi - 2) = 2\pi a^2$, exactly half of the surface of the ball of radius a. This is one more independent confirmation that the region drawn in Figure 2 is the region used in the integral.

The two figures together confirm the region used in part (a). The projection in Figure 2 is the disc whose area is $\pi a^2/4$; the height of the solid above that disc is the sphere height $\sqrt{a^2 - r^2}$; and the integral (7) is exactly the integral of that height over that disc.

How to solve problems of this kind

1. Convert the cylindrical or spherical surface equation to Cartesian form and identify the surface. Here $r = a\cos\theta$ becomes $(x - a/2)^2 + y^2 = (a/2)^2$: a cylinder, not a sphere. Knowing the axis and the radius prevents most drawing and range errors.
2. Find the range of the angle from the requirement $r \geq 0$. A factor $\cos\theta$ forces $|\theta| \leq \pi/2$. Use one full period and keep the range where the radial limit is non-negative.
3. Find the z-limits from the other surface. For a sphere, solve for z and get $z = \pm\sqrt{a^2 - r^2}$. Check that the square root is real on the whole projection; if it is not, the projection is too large.
4. Check that the region is simple over the projection, then choose the order of integration. If every vertical line meets the solid in one segment, dz goes innermost and $dV = r\,dz\,dr\,d\theta$.
5. Integrate from the inside out. When an antiderivative produces $\sqrt{\sin^2\theta}$, replace it by $|\sin\theta|$ and use symmetry.
6. Check the answer: compare with the volume of the enclosing ball, run a quadrature and a Monte Carlo estimate, and look for a classical formula for the same shape.